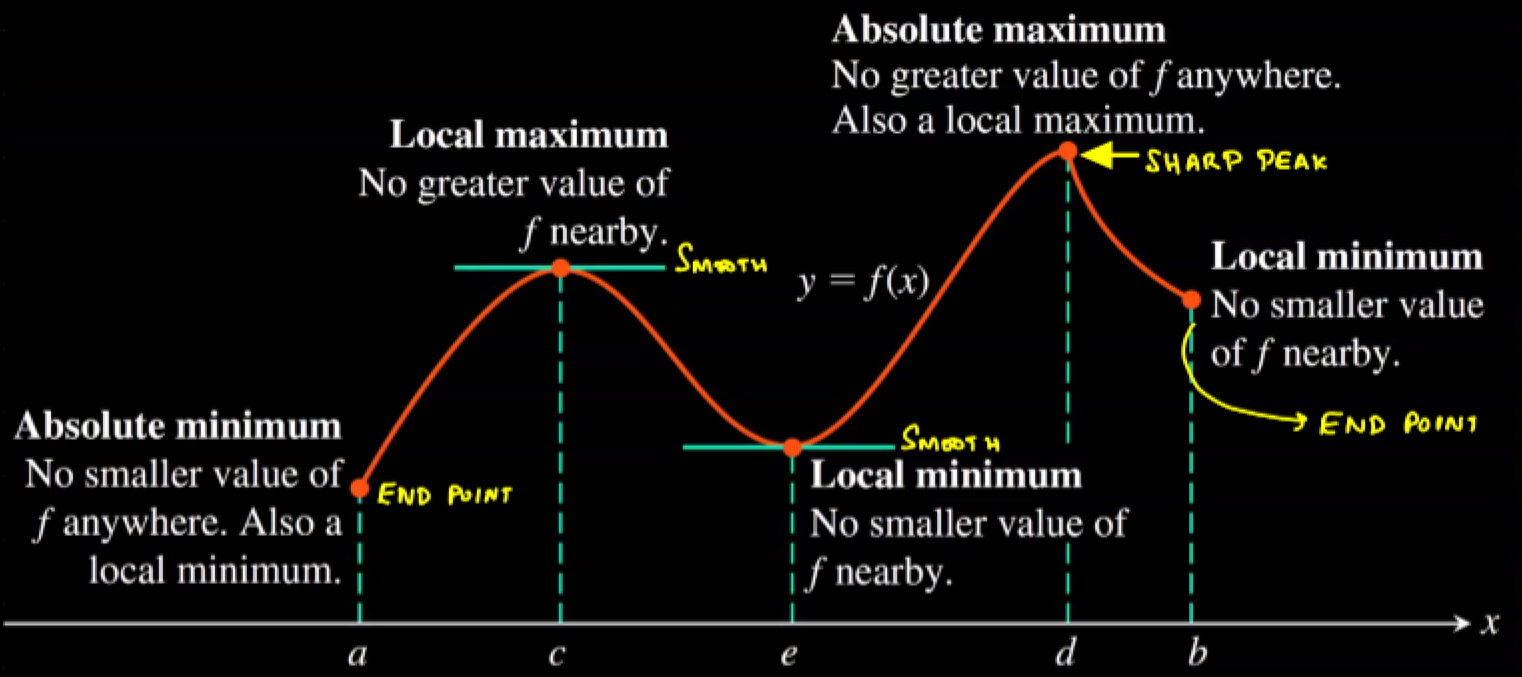
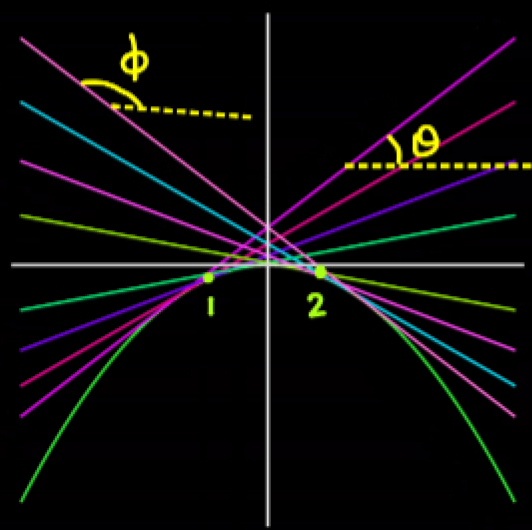
MAXIMA AND MINIMA:



The figure above shows various types of maxima or minima that we come across while analysing the functions. In physics, most of the times we are interested in locating the smooth peaks and smooth valleys (smooth = no sudden change in slope). For such peaks and valleys we can easily see that tangent becomes parallel to the horizontal axis and hence slope becomes zero and hence it can be located by setting the derivative to zero. The maxima and minima are combinedly called extrema.

LOCATING "SMOOTH" MAXIMA:

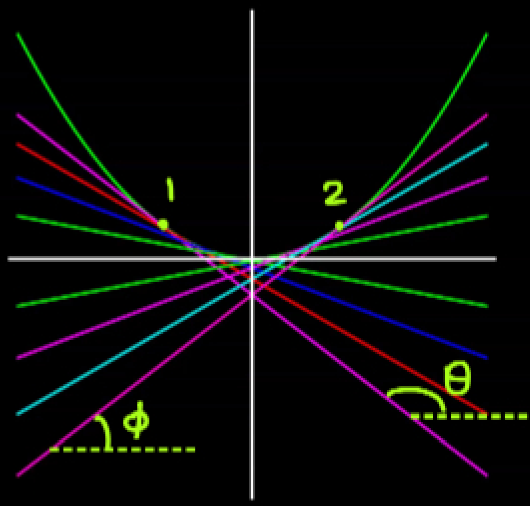


Consider the behaviour of a function near its peak, as one moves from point 1 to point 2 through the peak, one can clearly see that, initially the upward tangent makes an acute angle with positive x axis (θ), right at the peak this angle becomes zero and then it becomes obtuse. Thus $\tan\theta$ is initially positive ($\because$ acute angle) and then becomes negative after passing through a zero at the peak. Also recall $\tan\theta$ represents dy/dx, thus rate of change of $\tan\theta$ represents $\frac{d}{dx}\left(\frac{dy}{dx}\right)$ or $\frac{d^2y}{dx^2}$, now since rate of change of $\tan\theta$ is -ve with increasing x, thus at the peak, $d^2y/dx^2 < 0$ and $dy/dx = 0$. Thus for a maximum:

$$\boxed{\frac{dy}{dx} = 0} \ \& \ \boxed{\frac{d^2y}{dx^2} < 0}$$

Locating "Smooth" Minima: :

Consider the behavior of a function near its valley. As one moves from point 1 to point 2 through the valley one can see that angle of upward tangent is initially obtuse (θ) and then it becomes acute (ϕ) after passing the valley (exactly at the valley it is zero). Thus the value of $\tan\theta$ becomes positive from negative upon crossing the valley. But $\tan\theta = dy/dx$, thus we can say that dy/dx is increasing with increase in x or $\frac{d}{dx}(dy/dx) > 0$ or $d^2y/dx^2 > 0$ Thus for a minimum:



$$\boxed{\frac{dy}{dx} = 0} \ \& \ \boxed{\frac{d^2y}{dx^2} > 0}$$

→ For finding maxima and minima in a limited domain, we must also check at the end points of the domain.

Example: Find the absolute maximum and minimum values of function $f(x) = 10x(2 - \ln x)$ on the interval $[1, e^2]$.

Solution: For critical points $f'(x) = 0 \Rightarrow 10x\left(\frac{-1}{x}\right) + (2 - \ln x) \times 10 = 0$
$\Rightarrow 2 - \ln(x) = 1 \Rightarrow \ln(x) = 1 \Rightarrow x = e.$

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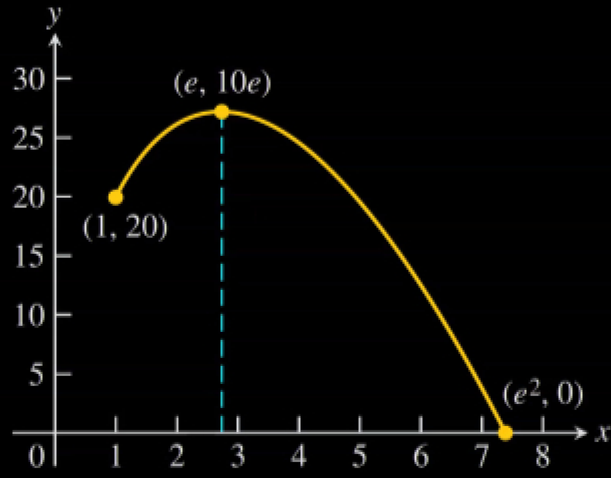
Now $f''(x) = \frac{-10}{x}$, $f''(e) = -\frac{10}{e} < 0 \Rightarrow x = e$ should be a point of maximum.



Now at domain end points:

$f(1) = 20$, $f(e^2) = 0$

and at critical point, $f(e) = 10e$

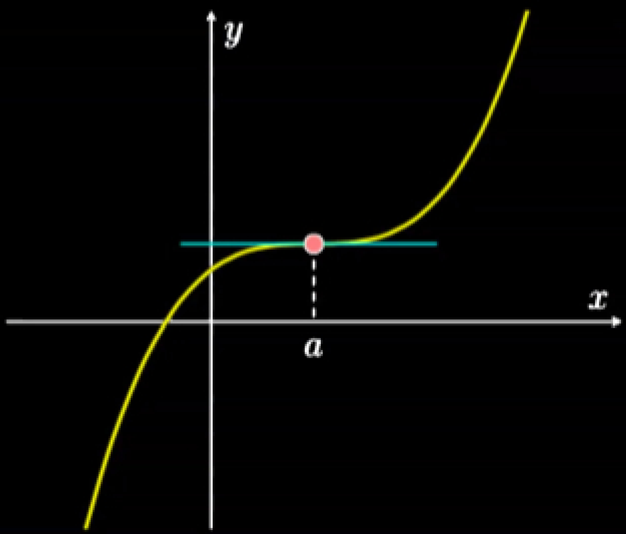
Thus absolute maxima: $10e$ for $x = e$

Absolute minimum: 0 for $x = e^2$

→ A point of zero derivative is called critical point.

Critical Point With Neither Maximum Nor Minimum:

There may exist points on a graph where tangent is parallel to x-axis however still it is neither a maximum or minimum. Such has point has different signs of second derivative on either side of the critical point and second derivative will be neither positive nor negative at such a point. For instance the function $f(x) = x^3$ has a critical point



at $x=0$ which is neither a maximum nor a minimum at $x=0$. Its second derivative is $6x$ which is negative slightly to the left of origin and positive slightly to the right of the origin.

Example: For following functions, locate critical points and comment whether they are maximum, minimum or neither:

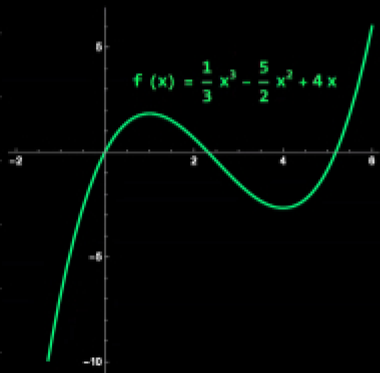
(a) $\frac{1}{3}x^3 - \frac{5}{2}x^2 + 4x$ (b) $(x^2-1)^3$ (c) $4x/(1+x)^2$

Solution: (a) $f'(x) = x^2 - 5x + 4 = (x-4)(x-1) = 0$

$\Rightarrow x = 4, 1$ $f''(x) = 2x - 5$, $f''(4) = 3 > 0$

$\Rightarrow x = 4$ must be minima, $f''(1) = -3 < 0 \Rightarrow$

$x = 1$ must be point of maximum. Ans.



(b) $f(x) = (x^2-1)^3 \Rightarrow f'(x) = 3(x^2-1)^2 \cdot 2x$

$\Rightarrow f''(x) = 6\,[x \times 2(x^2-1)\cdot 2x + (x^2-1)^2 \cdot \times 1]$

$\Rightarrow f''(x) = 24x^2(x^2-1) + (x^2-1)^2$

Now putting $f'(x) = 0$, $6x(x^2-1)^2 = 0$

$\Rightarrow x = 0, 1, -1$

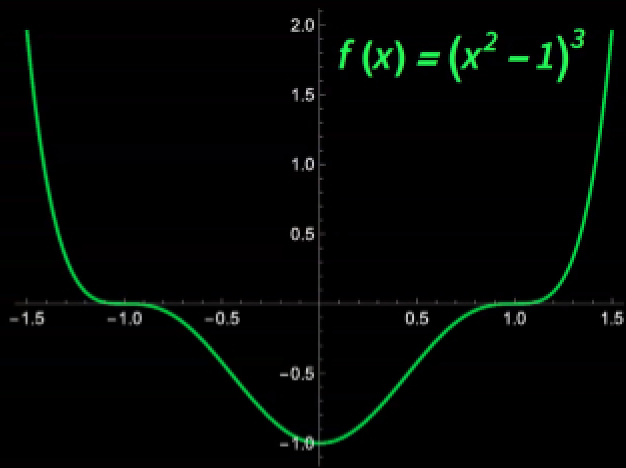
Now $f''(0) = +1 \Rightarrow x = 0$ is a minimum.

$f''(1) = 0$, Also $f''(x) = (x^2-1)[25x^2 - 1]$

Now $f''(1_-) < 0$ and $f''(1_+) > 0$

Thus $x = 1$ is neither maximum nor minimum. Similarly, $f''(-1) = 0$, $f''(-1_-) > 0$, $f''(-1_+) < 0$

Thus $x = -1$ is also neither maximum nor minimum.

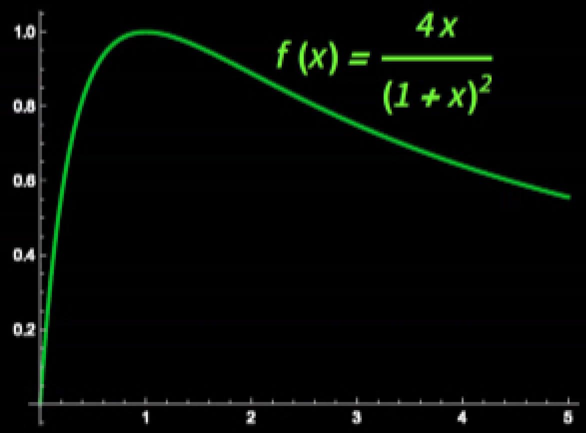


(c) $f(x) = 4x/(1+x)^2$,

$f'(x) = 4(x-1)/(x+1)^3$, Putting $f'(x) = 0$, $x = 1$.

Now $f''(x) = \dfrac{8(x-2)}{(x+1)^4}$, $f''(1) < 0$

$\Rightarrow x = 1$ is a point of maximum.

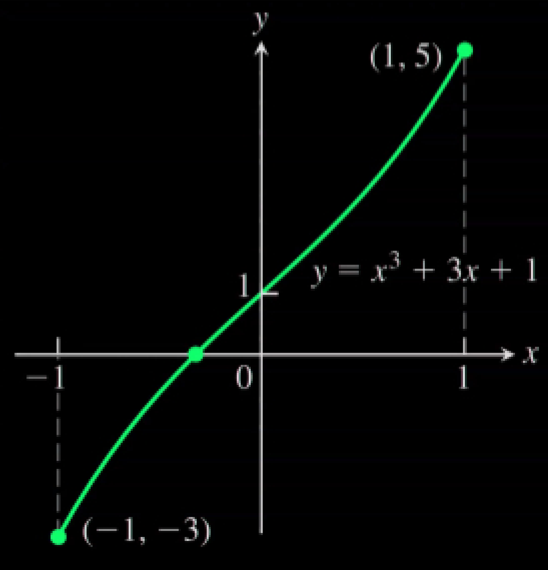


→ Btw domain of $f(x)$ is $\mathbb{R} - \{-1\}$

($\mathbb{R}$ represents set of all real numbers, the function is defined for all x except $x = -1$), $\mathbb{R} - \{-1\} = (-\infty, -1) \cup (-1, \infty)$

EXAMPLE: Show that Equation: $f(x) = x^3 + 3x + 1 = 0$ has exactly one solution.

Solution: $f(-\infty) < 0$, $f(\infty) > 0$, Since the function is changing its sign, it must cut the x axis somewhere, thus there should be atleast one root. Now, $f'(x) = 3x^2 + 3$ which always > 0. Thus once the function cuts x axis it will keep rising without ever going down to cut x-axis again. Thus the function has exactly one root.

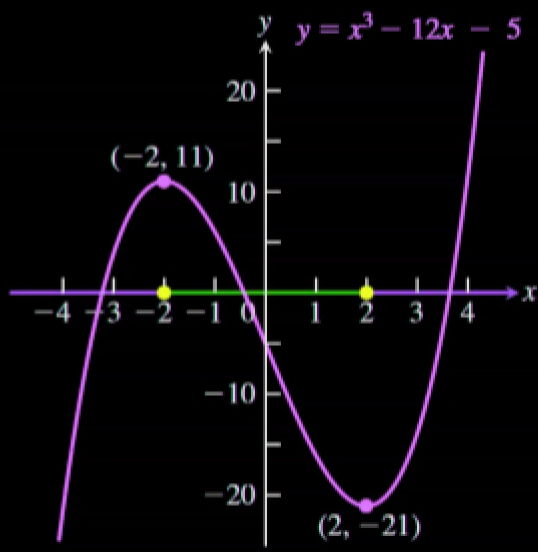


INCREASING AND DECREASING FUNCTIONS: If $f'(x) > 0 \ \forall x \in [a,b]$ then $f(x)$ is monotonically increasing on the interval $[a,b]$ (i.e it keeps increasing without ever decreasing), Similarly if $f'(x) < 0 \ \forall x \in [a,b]$ then $f(x)$ is monotonically decreasing on the interval $[a,b]$.

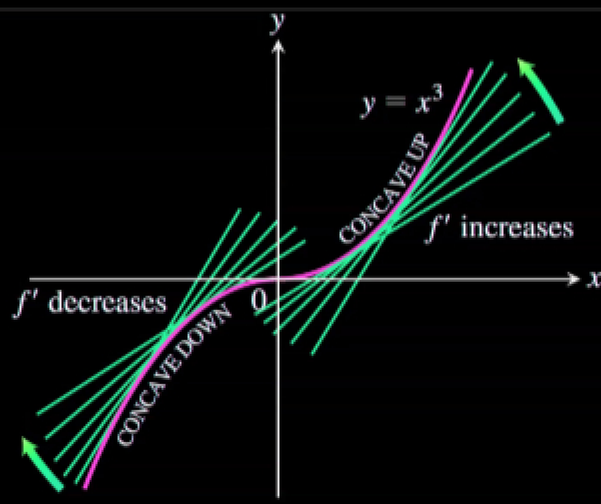
EXAMPLE: Find critical points for $f(x) = x^3 - 12x - 5$ and identify intervals for which it is increasing and decreasing.

Solution: $f'(x) = 3x^2 - 12$, $f'(x) = 0 \Rightarrow x = \pm 2$

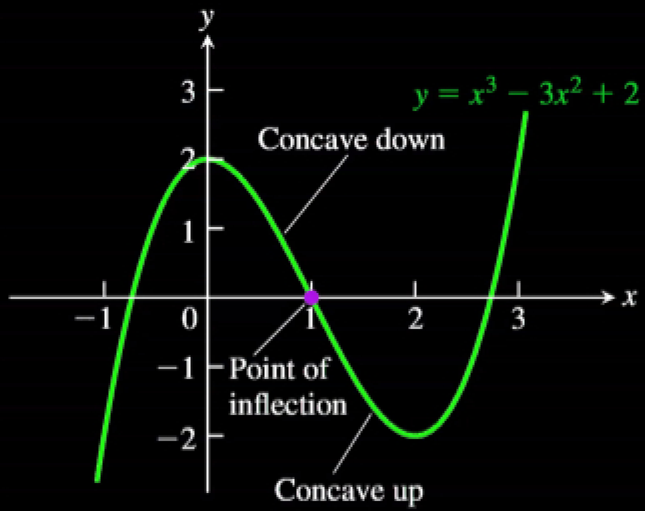
Whenever $f'(x) = 0$ there the function is possibly changing its nature from increasing to decreasing or vice-versa. We can see that $f'(x) > 0$ for $x \in (-\infty, -2)$ and $x \in (2, \infty)$ thus it is increasing in the intervals $(-\infty, -2) \cup (2, \infty)$ and decreasing in the interval $(-2, 2)$. Ans.



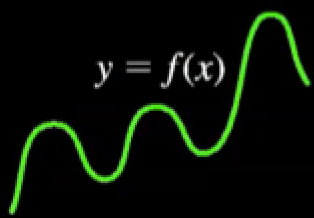
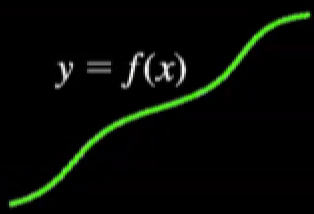
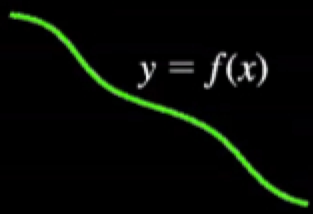
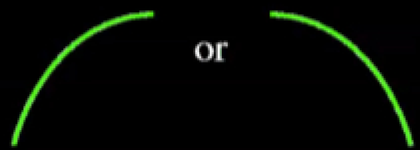
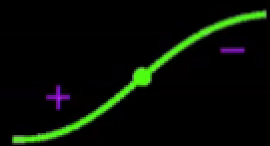
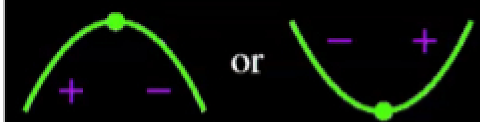
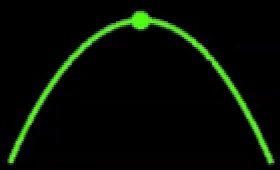
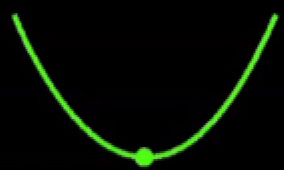
CONCAVITY: If $f'(x)$ increases in an interval the curve is called concave up thus for concave up, $f''(x) > 0$ similarly if $f''(x) < 0$, the curve is concave down. Refer to adjacent figure for more clarity. Thus just as first derivative has geometrical significance of steepness (greater $|f'(x)|$ steeper the graph), the second derivative of the curve tells us about its concavity (or how fast and in which sense is the tangent of the curve turning).



Point Of Inflection: A point where a curve changes its concavity sense (from upward to downward or other way round) is called point of inflection. $f''(x)$ changes sign as one crosses point of inflection. For $f(x) = x^3 - 3x^2 + 2x$, $f'(x) = 3x^2 - 6x$, $f''(x) = 6x - 6$, this is zero at $x = 1$ and before that it is negative and after that it is +ve thus $x = 1$ is point of inflection.

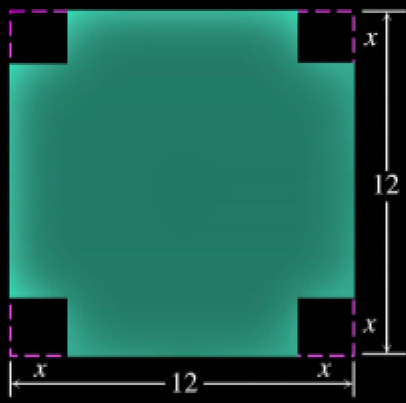


Graphical Behaviour Of Functions From Derivatives:

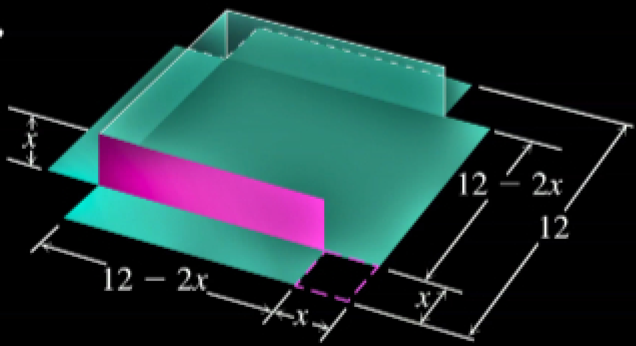
 Differentiable $\Rightarrow$ smooth, connected; graph may rise and fall	 $y' > 0 \Rightarrow$ rises from left to right; may be wavy	 $y' < 0 \Rightarrow$ falls from left to right; may be wavy
 $y'' > 0 \Rightarrow$ concave up throughout; no waves; graph may rise or fall or both	 $y'' < 0 \Rightarrow$ concave down throughout; no waves; graph may rise or fall or both	 y'' changes sign at an inflection point
 y' changes sign $\Rightarrow$ graph has local maximum or local minimum	 $y' = 0$ and $y'' < 0$ at a point; graph has local maximum	 $y' = 0$ and $y'' > 0$ at a point; graph has local minimum

APPLIED OPTIMIZATION:

An open-top box is to be made by cutting small congruent squares from the corners of a 12cm x 12cm sheet of tin and bending up the sides. How large should the squares cut from the corners be to make the box hold as much as possible?



<u>Solution:</u> Here we will try to express the volume as a function of a general variable x and then try to maximize the Volume function $V(x)$ wrt to the variable x. Consider the adjacent figure.



$$V = (12-2x)^2 \times x = 4x^3 - 48x^2 + 144x$$

Now for extremum, $V'(x) = 0$

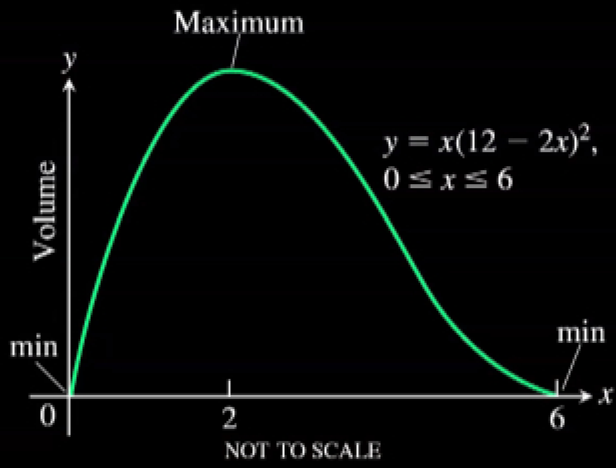
$\Rightarrow 12x^2 - 96x + 144 = 0$

$\Rightarrow 12(x^2 - 8x + 12) = 0 \Rightarrow (x-6)(x-2) = 0$

$\Rightarrow x = 6, 2.$

Now $V''(x) = 24x - 96$,

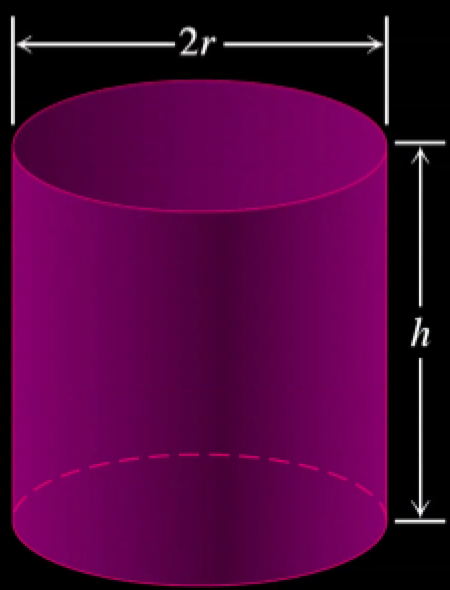
Now $V''(6) > 0$ thus it corresponds to minimum, and $V'(2) < 0 \Rightarrow$ it corresponds to maximum. Thus max volume occurs at $x = 2$ and max volume is $V(2) = 128 \text{ cm}^3$ Ans.



EXAMPLE: You have to design a cylindrical Can with lid so as to maximize its capacity. What should be the ratio h/r to achieve this.

<u>Solution:</u> Let required capacity be V and the ratio $h/r = x$. Then,

$$V = \pi r^2 \times xr = \pi r^3 x \quad (1)$$



Now surface area:

$$S = 2\pi r \times xr + \pi r^2 \times 2$$

$$\Rightarrow \quad S = 2\pi r^2 [x+1] \quad (2)$$

From (1), $r = (V/\pi x)^{1/3}$

$$\Rightarrow \quad S = 2\pi x\left(\frac{V}{\pi x}\right)^{2/3} \cdot [x+1] = 2\pi^{1/3} V^{2/3} x^{-2/3}(x+1)$$

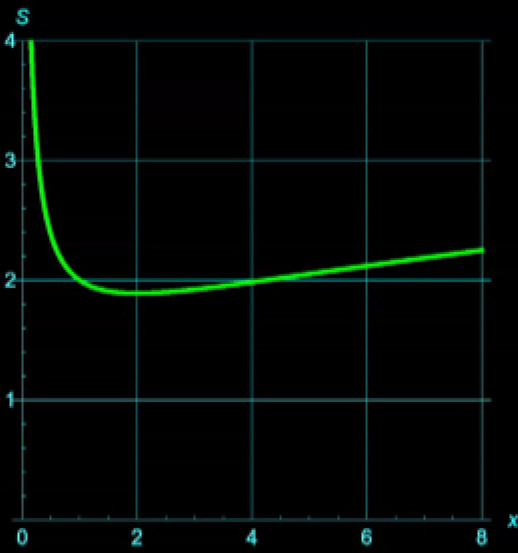
Now for minimizing S we need to minimize $\dfrac{x+1}{x^{2/3}}$ or $\dfrac{(x+1)^3}{x^2} = E$ (say)

Now $E = (x+1)^3/x^2 = (1/x^2)(x^3+3x^2+3x+1) = x+3+\frac{3}{x}+1/x^2$.

$$\frac{dS}{dx} = 0 \Rightarrow 1 - \frac{3}{x^2} - \frac{2}{x^3} = 0, \Rightarrow x^3 - 3x - 2 = 0 \Rightarrow (x+1)^2(x-2) = 0$$

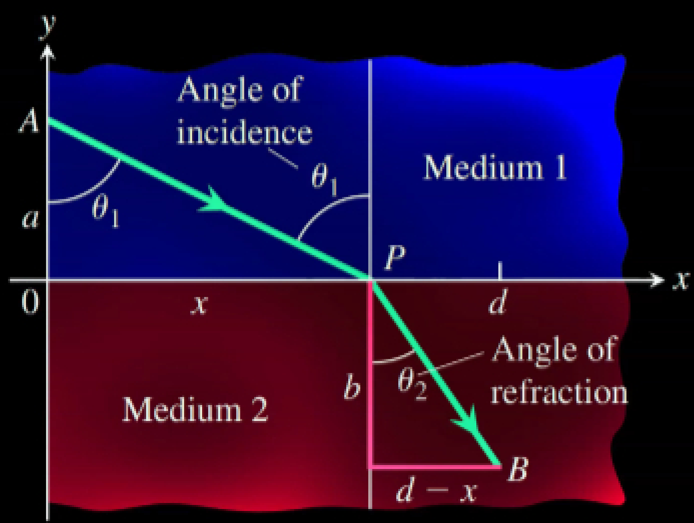
Thus our required root must be $x = 2$

$\therefore$ For best can, $h/r = 2$ Ans.



The adjacent figure gives the plot of Surface area with x for $2\pi^{1/3}V^{2/3} = 1$. If the ratio is made smaller than $1/2$ we will have taller and narrow can and if x is greater than 2 we get a short and wide can, both with larger surface areas.

EXAMPLE: Consider the adjacent figure, it is desired to go from point $A(0,a)$ to point $B(d,-b)$. The speed in medium 1 is c/μ_1 and that in medium 2 is c/μ_2. Find the relation between θ_1 and θ_2 for quickest path between A and B.



Solution: Let T be time from A to B then:

$$T = \frac{a\sec\theta_1}{c/\mu_1} + \frac{b\sec\theta_2}{c/\mu_2} = \frac{1}{c}\left[\mu_1 a\sec\theta_1 + \mu_2 b\sec\theta_2\right] \quad (1)$$

Also: $a\tan\theta_1 + b\tan\theta_2 = d$ (2)

Differentiating (1) wrt θ_1 (For minimum, $dT/d\theta_1 = 0$)
we get:

$$\mu_1 a \sec\theta_1 \tan\theta_1 + \mu_2 b \sec\theta_2 \tan\theta_2 \cdot \frac{d\theta_2}{d\theta_1} = 0 \quad (3)$$

Now Differentiating (2) $a\sec^2\theta_1 + b\sec^2\theta_2 \dfrac{d\theta_2}{d\theta_1} = 0$ (4)

Now eliminating $d\theta_2/d\theta_1$ from (3) & (4):

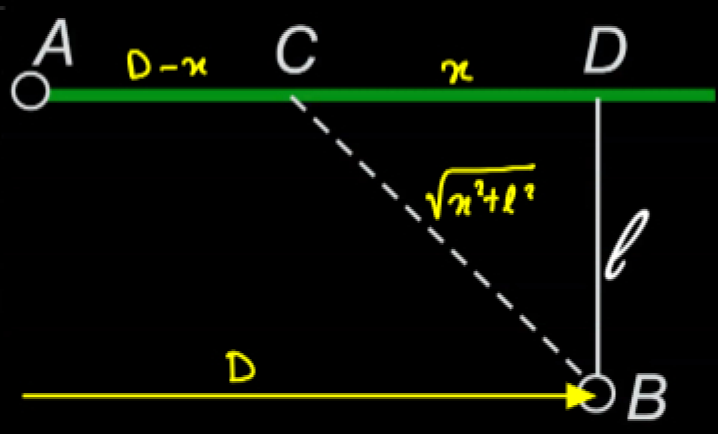
$$\frac{\mu_1 a \sec\theta_1 \tan\theta_1}{a\sec^2\theta_1} = \frac{-\mu_2 b \sec\theta_2 \tan\theta_2}{-b\sec^2\theta_2}\left(\frac{d\theta_2/d\theta_1}{d\theta_2/d\theta_1}\right)$$

$\Rightarrow \mu_1 \sin\theta_1 = \mu_2 \sin\theta_2$ Ans.

→ We just derived snell's law using fermat's principle).

EXAMPLE: IRODOV 1.17:

From point A located on a highway (see Fig.) one has to get by car as soon as possible to point B located in the field at a distance ℓ from the highway. It is known that the car moves in the field η times slower than on the highway. At what distance from point D one must turn off the highway?



Solution: Let the speed in the field be v then on highway, speed is ηv.
Now let T be the time to reach B then:

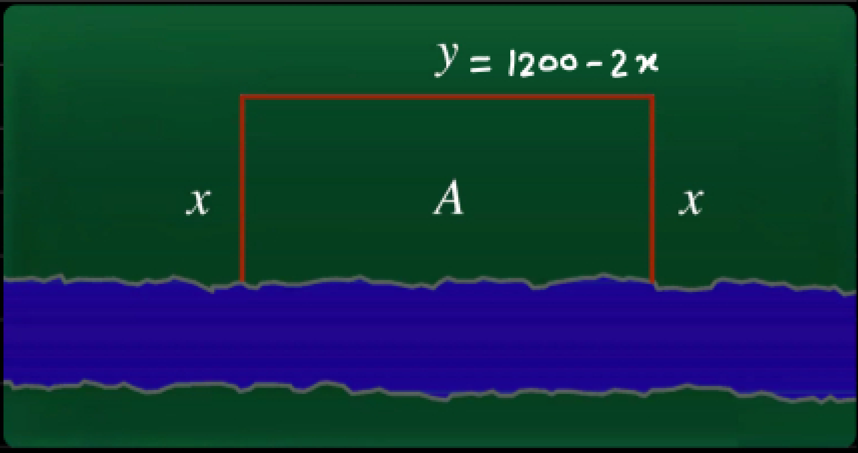
$$T = \frac{D-x}{\eta v} + \frac{\sqrt{x^2+\ell^2}}{v} \qquad \text{Now for shortest time, } dT/dx = 0$$

$$\Rightarrow \frac{1}{v}\left[\frac{-1}{\eta} + \frac{1}{2\sqrt{\ell^2+x^2}}\cdot 2x\right] = 0 \Rightarrow \frac{x}{\sqrt{\ell^2+x^2}} = \frac{1}{\eta}$$

$$\Rightarrow \eta^2 x^2 = \ell^2 + x^2 \Rightarrow (\eta^2-1)x^2 = \ell^2 \Rightarrow x = \ell/\sqrt{\eta^2-1} \text{ Ans.}$$

EXAMPLE:

A farmer has 1200 m of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?



Solution: Here we express the area of the field in terms of a general variable x. Thus: $A(x) = x(1200 - 2x) = 1200x - 2x^2$.

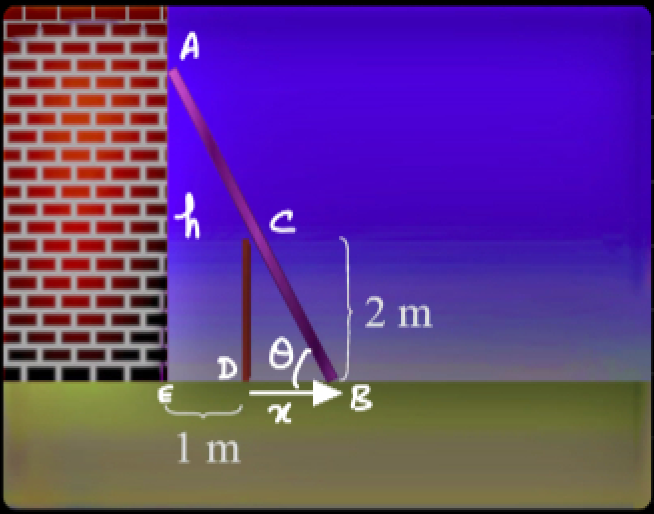
Now for maximum area, $A'(x) = 0 \Rightarrow 1200 - 4x = 0 \Rightarrow x = 300$

Also $A''(x) = -4 < 0$ thus $x = 300$ must be the maximum point.

Thus required dimensions: 300×600 Ans.

EXAMPLE:

A fence 2 m tall runs parallel to a tall building at a distance of 1 m from the building. What is the length of the shortest ladder that will reach from the ground over the fence to the wall of the building?



Solution: $\tan\theta = 2/x$ ① (x is the distance DB).

Now length of ladder: $l = (1+x)\sec\theta = (1+x)\sqrt{1+\tan^2\theta}$

$\Rightarrow l = (1+x)\sqrt{1+4/x^2} = \left(\frac{x+1}{x}\right)\sqrt{x^2+4}$

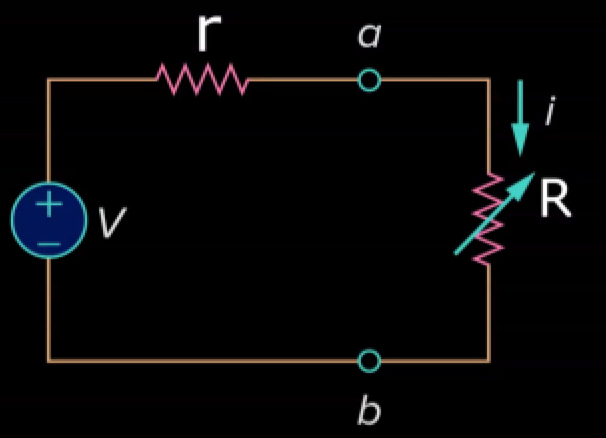
Now for min l, $\frac{dl}{dx} = 0 \Rightarrow \left(1+\frac{1}{x}\right)\cdot\frac{1}{2\sqrt{x^2+4}}\cdot 2x + \sqrt{x^2+4}\cdot\left[-\frac{1}{x^2}\right] = 0$

$\Rightarrow x^3\left[1+\frac{1}{x}\right] = x^2+4 \Rightarrow x^3+x^2 = x^2+4 \Rightarrow x = 4^{1/3}$

Putting this in l, $l = (1+4^{1/3})^{3/2} \simeq 4.16$ m Ans.

EXAMPLE: Power dissipated in an external variable resistor due to a cell of EMF V and internal resistance r is given as $P = \frac{V^2 R}{(R+r)^2}$

For what value of R is the power dissipation maximized.



Solution: For max Power, $dP/dR = 0$

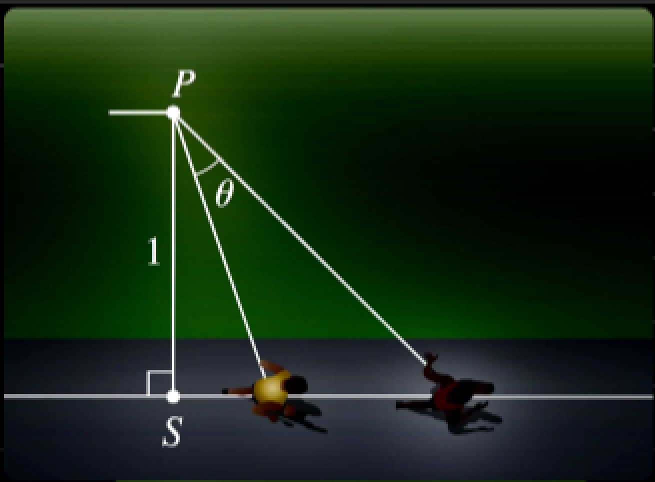
$$\Rightarrow V^2\left[\frac{(R+r)^2 \times 1 - R \cdot 2(R+r)\times 1}{(R+r)^4}\right] = 0 \Rightarrow R^2 + 2Rr + r^2 = 2R^2 + 2Rr$$

$\Rightarrow R = r$ Ans. Thus for maximum power dissipation external resistance must be equal to internal resistance.

→ We just derived maximum power transfer theorem.

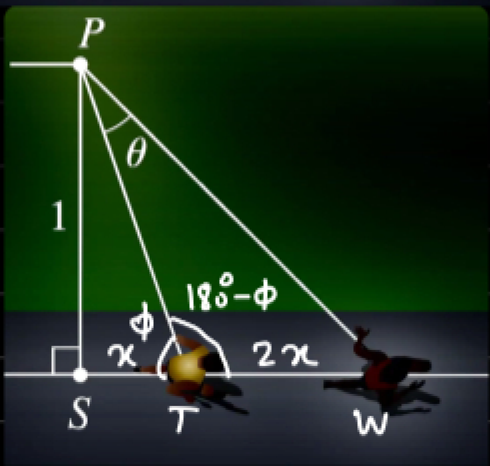
EXAMPLE:

An observer stands at a point P, one unit away from a track. Two runners start at the point S in the figure and run along the track. One runner runs three times as fast as the other. Find the maximum value of the observer's angle of sight θ between the runners.



Solution: Using Sine law:

$$\frac{2x}{\sin\theta} = \frac{\sqrt{1+(3x)^2}}{\sin(180^\circ - \phi)} \quad (1)$$



Also $\sin\phi = \frac{1}{\sqrt{1+x^2}}$ (2)

Now from (1), $\sin\theta = \frac{2x\sin\phi}{\sqrt{1+9x^2}}$

$$\Rightarrow \sin\theta = \frac{2x}{\sqrt{1+9x^2} \times \sqrt{1+x^2}}$$

Now to maximize $\sin\theta$ we need to maximize RHS. Let square of RHS be q

$\Rightarrow q = \frac{4x^2}{(1+x^2)(1+9x^2)}$, or we can also mimize $1/q$.

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$$\frac{1}{q} = \frac{(1+x^2)(1+9x^2)}{4x^2} = \frac{1}{4}\left[\left(1+\frac{1}{x^2}\right)\left(1+9x^2\right)\right]$$

$$\Rightarrow \frac{1}{q} = \frac{1}{4}\left[1+9x^2+\frac{1}{x^2}+9\right] = \frac{1}{4}\left[10+\frac{1}{x^2}+9x^2\right]$$

Differentiating wrt x and equating to zero:

$$\frac{-2}{x^3} + 18x = 0 \Rightarrow 18x^4 = 2 \Rightarrow x = \frac{1}{\sqrt{3}}$$

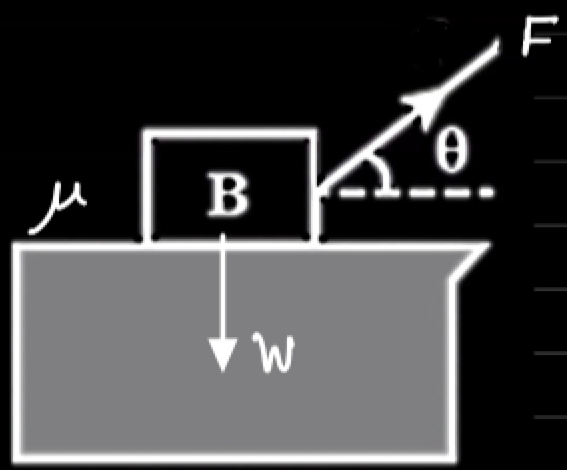
Evaluating $\sin\theta$ for this x: $\sin\theta = 1/2 \Rightarrow \theta = 30^\circ$ Ans.

EXAMPLE:

An object with weight W is dragged along a horizontal plane by a force acting along a rope attached to the object. If the rope makes an angle θ with a plane, then the magnitude of the force is

$$F = \frac{\mu W}{\mu \sin\theta + \cos\theta}$$

where μ is a constant called the coefficient of friction. For what value of θ is F smallest?



Solution: To minimize F we might as well maximize its denominator. Now differentiating denominator wrt θ and equating to zero:

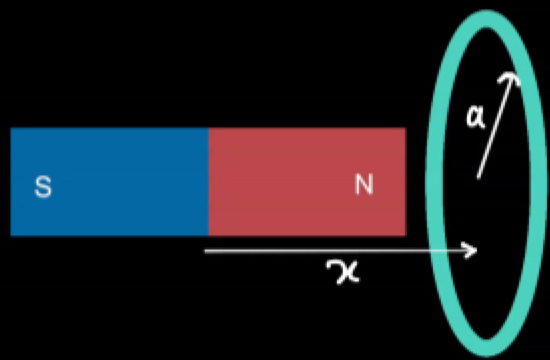
$$\mu\cos\theta - \sin\theta = 0 \Rightarrow \tan\theta = \mu \text{ or } \theta = \tan^{-1}\mu \text{ Ans.}$$

EXAMPLE: Force of circular current acting on a small magnet with its axis perpendicular to plane of circle and passing through its centre is given by the formula

$$F = \frac{Cx}{(a^2+x^2)^{3/2}}, \text{ where :}$$

a: radius of circle, x: distance of magnet

C: Constant. For what value of x will the force F be maximum?

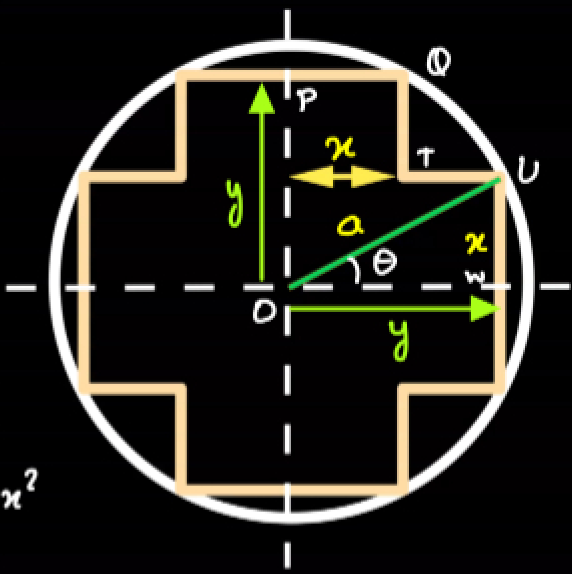


Solution: $$F'(x) = C\left[\frac{(a^2+x^2)^{3/2}\times 1 - x\times\frac{3}{2}(a^2+x^2)^{1/2}\times 2x}{(a^2+x^2)^3}\right] = C\,\frac{a^2-2x^2}{(a^2+x^2)^{5/2}} = 0$$

$\Rightarrow x = a/\sqrt{2}$ Ans. (There is only one root so this must be maximum).

→ Very often in physics problems it is unnecessary to mathematically check for maximum or minimum, because we are able to infer it from purely physics arguments based on physical situation.

EXAMPLE: In constructing an A.C. Transformer we need to insert a cross shaped iron core of maximum possible cross sectional area. Find most suitable x and y if radius of coil is a.



Solution: $a^2 = x^2 + y^2$ (1)

Now Area PQTUWOP can be found by breaking in two two rectangles, $A = xy + x(y-x) = 2xy - x^2$

Now to maximise, $dA/dx = 0$

$\Rightarrow 2x\frac{dy}{dx} + 2y - 2x = 0$ (2)

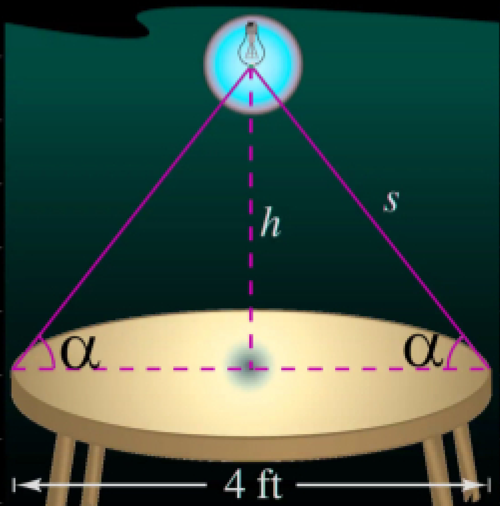
Differentiating (1) wrt x: $0 = 2x + 2y\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$ (3)

Using (3) in (2), $-2\frac{x^2}{y} + 2y - 2x = 0$ (4)

Solving (1) & (4), $x = a\sqrt{\frac{5-\sqrt{5}}{10}}$, $y = a\left(\sqrt{1 - \frac{5-\sqrt{5}}{10}}\right)$

→ An alternate (easier) solution can be obtained by setting $x = a\sin\theta$ and $y = a\cos\theta$. The answer we get from that corresponds to $\theta = \frac{1}{2}\tan^{-1}(2)$.

EXAMPLE: A light source is located above centre of a circular table of diameter 4 ft. The illumination at the periphery of table is given as $I = k\sin\alpha/s^2$ where k is a constant and s and α are shown in the figure. Find h for which illumination is maximum.



Solution: $I = k\sin\alpha/s^2 = k \times \frac{h}{\sqrt{h^2+4}} \cdot \frac{1}{(h^2+4)}$

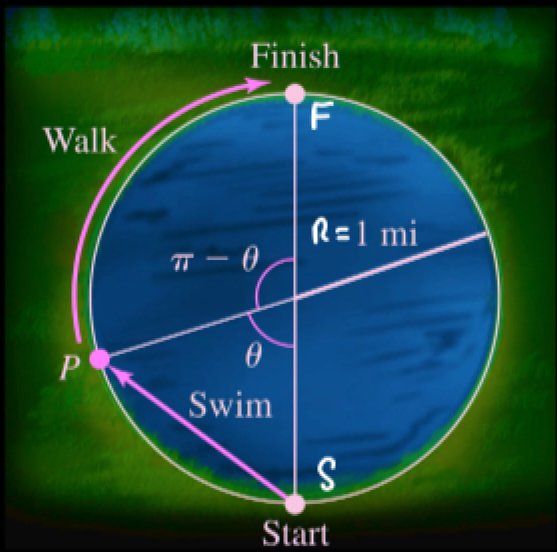
Thus $I \propto h(h^2+4)^{-3/2}$, For max I, $dI/dh = 0$

$\Rightarrow h \times -\frac{3}{2}(h^2+4)^{-5/2} \times 2h + (h^2+4)^{-3/2} \times 1 = 0$

Now multiplying throughout by $(h^2+4)^{5/2}$,

$-3h^2 + h^2 + 4 = 0 \Rightarrow 2h^2 = 4 \Rightarrow h = \sqrt{2}$ ft Ans.

EXAMPLE: You wish to reach from start point to finish point via a point P such that from start to P you swim and then you walk around pool. Swimming speed is 2 miles/hr and walking speed is 3 miles/hr. Find θ such that time from start to finish is least.



Solution: Time $T = \frac{2R\sin(\theta/2)}{v_s} + \frac{R(\pi-\theta)}{v_w}$

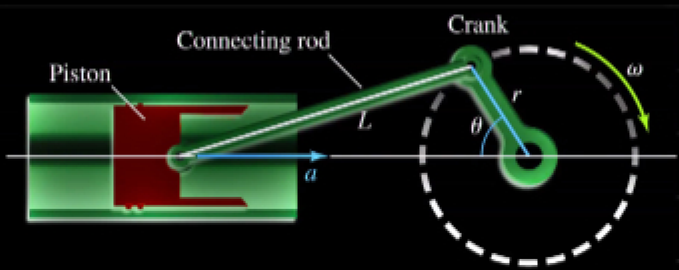
where v_s and v_w are swimming and walking speed respectively

Thus:

$$T = \frac{2\sin(\theta/2)}{2} + \frac{1(\pi-\theta)}{3} \quad \text{Now } \frac{dT}{d\theta} = 0$$

$\Rightarrow \frac{1}{2}\cos(\theta/2) - \frac{1}{3} = 0 \Rightarrow \cos(\theta/2) = 2/3 \Rightarrow \theta = 2\cos^{-1}(2/3) \approx 96^\circ$.

EXAMPLE: A crank of radius r rotates with an angular frequency ω. It is connected to a piston by a connecting rod of length L. The acceleration of the piston varies with position of the crank as $a(\theta) = \omega^2 r\left(\cos\theta + \frac{r\cos 2\theta}{L}\right)$, for fixed ω and r, find the values of θ for which acceleration is maximum and minimum. ($0 < \theta < 2\pi$).



Solution: For maximum/minimum acceleration, $da/d\theta = 0$

Now differentiating a w.r.t θ, $\frac{da}{d\theta} = \omega^2 r\left[-\sin\theta - \frac{2r}{L}\sin 2\theta\right] = 0$

Thus for critical points: $\sin\theta + \frac{4r}{L}\sin\theta\cos\theta = 0$ $(\because \sin 2\theta = 2\sin\theta\cos\theta)$

$\Rightarrow$ either $\sin\theta = 0$ or $\cos\theta = \frac{-L}{4r}$, for first case, critical points are $\theta = 0, \pi$ and 2π and second case, $\theta = \cos^{-1}\left(\frac{-L}{4r}\right)$ or $2\pi - \cos^{-1}\left(\frac{-L}{4r}\right)$ (L must be less than $4r$).

Now $a(0) = \omega^2 r\left[1 + \frac{r}{L}\right]$ $a(\pi) = \omega^2 r\left[-1 + \frac{r}{L}\right]$

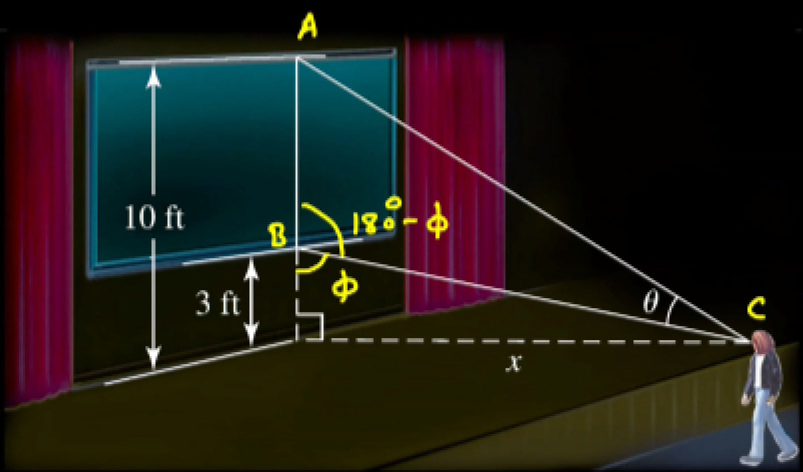
$a(2\pi) = \omega^2 r\left[1 + \frac{r}{L}\right]$, Now $a = \omega^2 r\left[\cos\theta + \frac{r}{L}\cos 2\theta\right]$

and $\cos 2\theta = 2\cos^2\theta - 1$ thus when $\cos\theta = -L/4r$,

$$a = \omega^2 r\left[\frac{-L}{4r} + \frac{r}{L}\cdot\left(2\frac{L^2}{16r^2} - 1\right)\right] = \omega^2 r\left[\frac{-L}{8r} - \frac{r}{L}\right]$$

Thus signed value maximum occurs at $\theta = 0$ & $\theta = 2\pi$ while minimum occurs at $\theta = \cos^{-1}(-L/4r)$ or $2\pi - \cos^{-1}(-L/4r)$ (Provided $L < 4r$, in that case $r/L > 4$ and $L/8r < 1/2$).

EXAMPLE: Consider the adjacent figure. It is desired to maximize the angle subtended on the eye for optimal viewing. Find the value of x.



Solution: Applying Sine law to the triangle ABC:

$$\frac{7}{\sin\theta} = \frac{\sqrt{10^2 + x^2}}{\sin(180^\circ - \phi)} \quad (1)\ (2)$$

Also $\sin\phi = \frac{x}{\sqrt{x^2 + 9}}$

Now from (1) & (2), $\sin\theta = \frac{7x}{\sqrt{x^2+9}} \cdot \frac{1}{\sqrt{100 + x^2}}$

Now to maximize $\sin\theta$, we need to minimize $\frac{1}{x^2}\left[(x^2+9)(x^2+100)\right]$

i.e minimize, $x^2 + \frac{900}{x^2} + 109 = y + \frac{900}{y} + 109$ where $y = x^2$

Now Differentiating wrt y and equating to zero:

$$1 - \frac{900}{y^2} = 0 \Rightarrow y^2 = 900 \Rightarrow y = 30 \Rightarrow x = \sqrt{30}\ \text{ft Ans.}$$

Thus required distance ≈ 5.5 ft Ans

RELATED RATES: Suppose a balloon is being filled with an air pump and we wish to know the rate of change of its radius when volume is increasing at a given rate, here if pumping rate is higher, the radius will be changing at a faster rate. Thus the rate of change of volume is related to rate of change of radius. The required relation can be established by differentiating the volume function $V(r)$ on both sides wrt time. Thus:

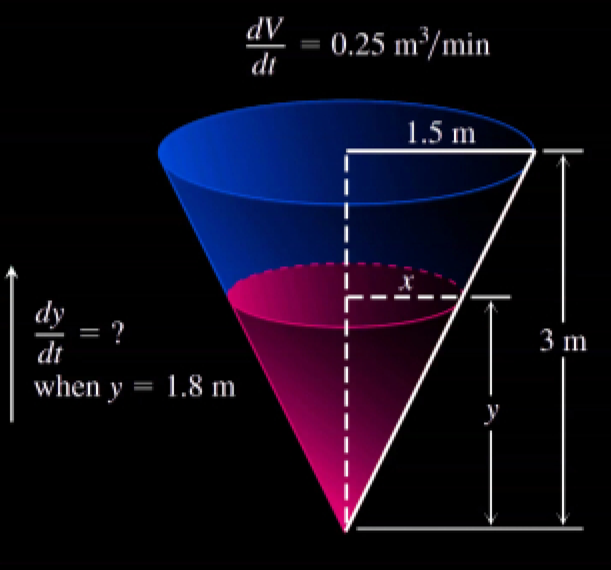
$$V = 4/3\,\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

or in Newton's notation: $\dot{V} = 4\pi r^2 \dot{r} \Rightarrow \dot{r} = \frac{1}{4\pi r^2}\dot{V}$

this is the required rate of change of radius.

EXAMPLE:

Water runs into a conical tank at the rate of $0.25\text{m}^3/\text{min}$. The tank stands point down and has a height of 3 m and a base radius of 1.5 m. How fast is the water level rising when the water is 1.8m deep?



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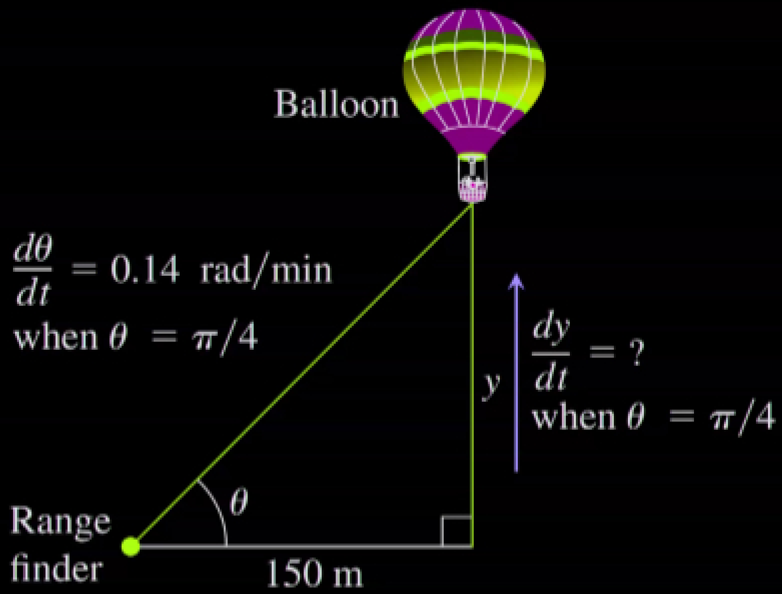
Solution: $V = \frac{1}{3}\pi x^2 y$ also $\frac{x}{y} = \frac{1.5}{3} \Rightarrow x = y/2$

$$\Rightarrow V = \frac{1}{12}\pi y^3 \Rightarrow \frac{dV}{dt} = \frac{\pi}{4} y^2 (dy/dt) \Rightarrow \dot{y} = \frac{4\dot{V}}{\pi y^2}$$

$$\Rightarrow \dot{y} = \frac{4 \times 0.25}{\pi \times 1.8^2} \approx 0.098 \text{ m/min Ans}$$

EXAMPLE :

A hot air balloon rising straight up from a level field is tracked by a range finder 150 m from the lift off point. At the moment the range finder's elevation angle is π/4, the angle is increasing at the rate of 0.14 rad/min. How fast is the balloon rising at that moment?

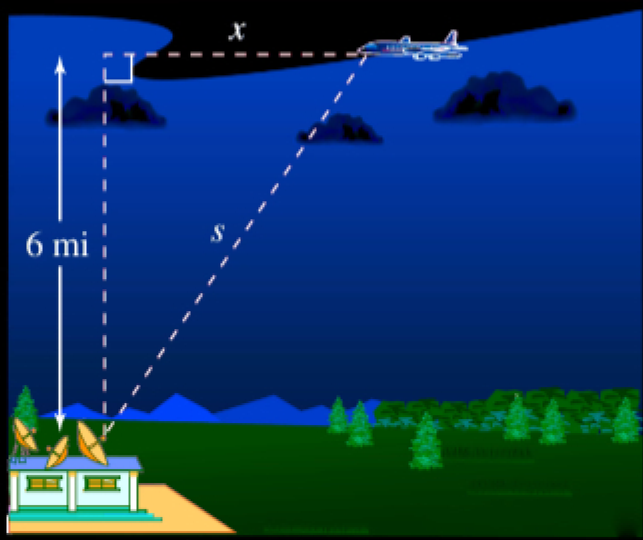


Solution: $y = 150 \tan\theta \Rightarrow \dot{y} = 150 \sec^2\theta\, \dot{\theta} = 150 \times \sec^2 45^\circ \times 0.14$

$$\Rightarrow \dot{y} \approx 42 \text{ m/min Ans.}$$

EXAMPLE :

An airplane is flying on a flight path that will take it directly over a radar tracking station, as shown in Figure. The distance s is decreasing at a rate of 400 miles per hour when s=10 miles. What is the speed of the plane?

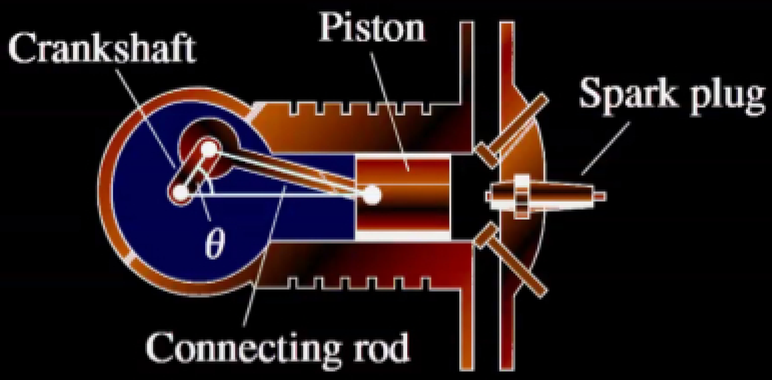


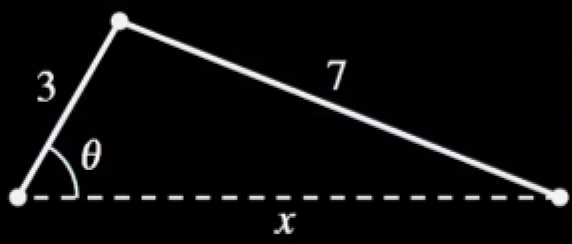
Solution: $x = \sqrt{s^2 - 6^2} \Rightarrow \dot{x} = \frac{1 \times 2s}{2\sqrt{s^2-36}} \times \dot{s}$

$$\Rightarrow \dot{x} = \frac{s}{\sqrt{s^2-36}} \cdot \dot{s} = \frac{10 \times (-400)}{\sqrt{100-36}} = -500 \text{ miles/hour. Ans.}$$

(−ve sign means x is decreasing with time).

EXAMPLE :





In the engine shown in Figure, a seven-inch connecting rod is fastened to a crank of radius 3 inches. The crankshaft rotates counter clockwise at a constant rate of 200 revolutions per minute. Find the velocity of the piston when $\theta = \pi/3$.

Solution: By cosine law: $x^2 + 3^2 - 6x\cos\theta = 7^2$ ①

$\Rightarrow 2x\dot{x} - 6\dot{x}\cos\theta + 6x\sin\theta\,\dot{\theta} = 0$

$\Rightarrow \dot{x}\,[2x - 6\cos\theta] = -6x\sin\theta\,\dot{\theta} \Rightarrow \dot{x} = \dfrac{-6\sin\theta\, x}{2(x - 3\cos\theta)}\,\dot{\theta}$ ②

Now when $\theta = \pi/3$, using ①, $x^2 - 3x - 40 = 0 \Rightarrow (x-8)(x+5) = 0$

$\Rightarrow x = 8$ inches, $\dot{\theta} = 200 \times 2\pi$ radians/min $= 400\pi$ radians/min.

$\Rightarrow$ using ②, $\dot{x} = \dfrac{-3\sin\pi/3 \times 8}{(8 - 3/2)} \times 400\pi$ inch/min

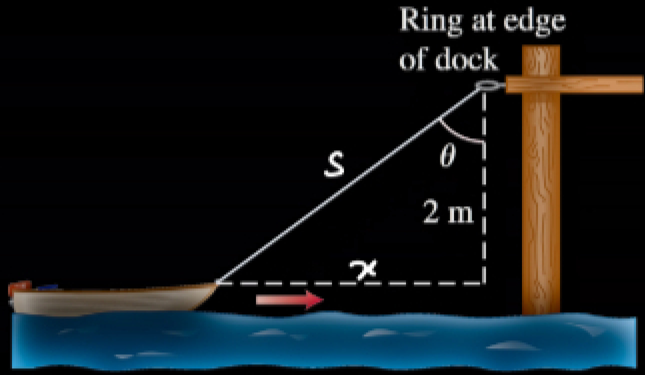
$\Rightarrow \dot{x} \simeq -4018$ in/min Ans. $\longrightarrow$ Always convert angle to radians.

EXAMPLE :

Hauling in a dinghy: A dinghy is pulled toward a dock by a rope from the bow through a ring on the dock 2m above the bow. The rope is hauled in at the rate of 0.5m/s.

a.) How fast is the boat approaching the dock when 3 m of rope are out?

b.) At what rate is the angle θ changing at this instant (see the figure)?



Solution: $x = \sqrt{s^2 - 4} \Rightarrow \dot{x} = \dfrac{1}{2\sqrt{s^2-4}} \cdot 2s \times \dot{s} = \dfrac{3 \times (-0.5)}{\sqrt{5}}$

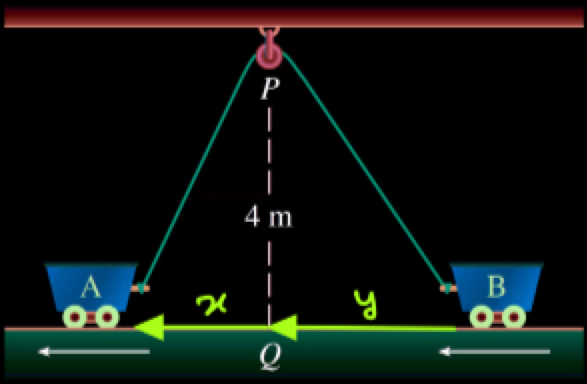
$\Rightarrow \dot{x} \simeq -0.67\ ms^{-1}$ Ans.

Now $\cos\theta = \dfrac{2}{s} \Rightarrow -\sin\theta\,\dot{\theta} = \dfrac{-2}{s^2}\dot{s} \Rightarrow \dot{\theta} = \dfrac{2}{s^2\sin\theta} \cdot \dot{s}$

$$= \frac{2}{9 \times (\sqrt{5}/3)} \times (-0.5) \approx -0.15 \text{ rad s}^{-1} \text{ Ans.}$$

EXAMPLE:

Two carts, A and B, are connected by a rope 12 m long that passes over a pulley P. (See the figure.) The point Q is on the floor 4 m directly beneath P and between the carts. Cart A is being pulled away from Q at a speed of 0.5 m/s. How fast is cart B moving toward Q at the instant when cart A is 3 m from Q?



Solution: $\sqrt{x^2+4^2} + \sqrt{y^2+4^2} = 12$ ①

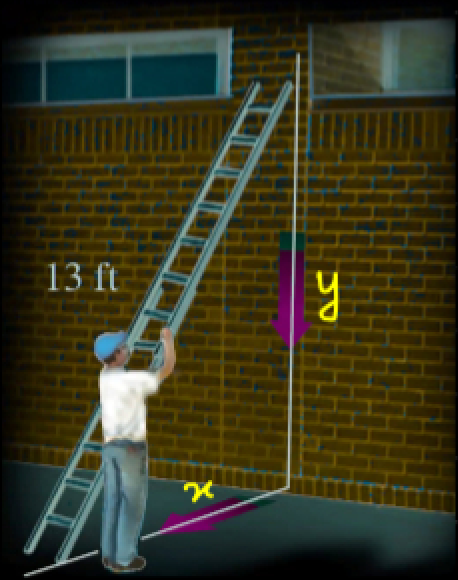
$$\Rightarrow \frac{x\,\dot{x}}{\sqrt{x^2+16}} + \frac{y}{\sqrt{y^2+16}}\,\dot{y} = 0 \Rightarrow \dot{y} = -\sqrt{\frac{y^2+16}{x^2+16}} \times \frac{x\dot{x}}{y}$$

Now at this moment $x = 3 \Rightarrow \sqrt{y^2+16} = 7$ ($\because$ ①) $\Rightarrow y = \sqrt{33}$.

Thus $\dot{y} = \frac{-3}{\sqrt{33}} \times 0.5 \times \frac{7}{5} \approx -0.37 \text{ m s}^{-1}$ Ans.

EXAMPLE:

Ladder against the wall: A 13-foot ladder is leaning against a vertical wall (see figure) when Jack begins pulling the foot of the ladder away from the wall at a rate of 0.5 ft/s. How fast is the top of the ladder sliding down the wall when the foot of the ladder is 5 ft from the wall?



Solution: $x^2+y^2 = 13^2 \Rightarrow 2x\dot{x} + 2y\dot{y} = 0$

$$\Rightarrow \dot{y} = -\frac{x}{y}\dot{x} = -\frac{5}{12} \times 0.5 \approx -2.1 \text{ ft/sec} \text{ Ans.}$$

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TAYLOR SERIES: If a function is infinitely differentiable at a real number x then:

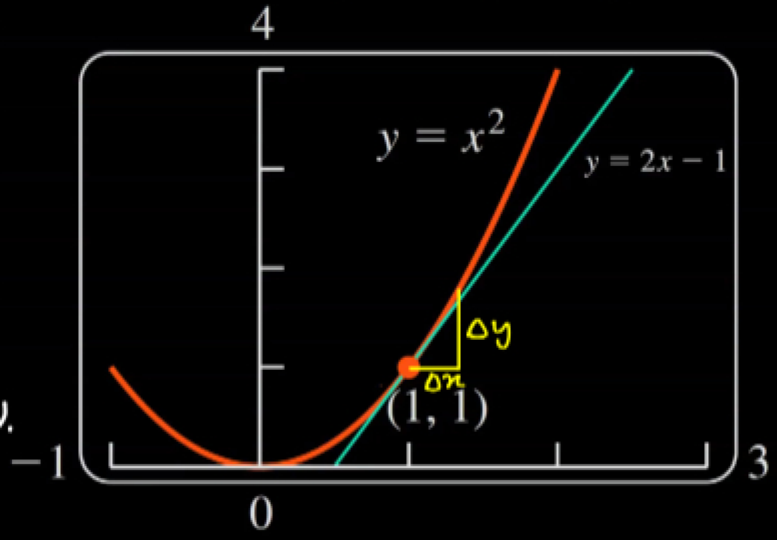
$$f(x+\Delta x) = f(x) + \frac{f'(x)\Delta x}{1!} + \frac{f''(x)\Delta x^2}{2!} + \frac{f'''(x)\Delta x^3}{3!} + \dots$$

TAYLOR APPROXIMATION: Depending on our requirement, we can consider as many terms of taylor series to get an approximate value of our function, however in many applications in physics it is sufficient to consider first order approximation i.e:

$$\boxed{f(x+\Delta x) \approx f(x) + f'(x)\Delta x}, \leftarrow$$ $f(x_0) + f'(x_0)(x-x_0)$ is equation of tangent.

Anothe way of stating it is:

$$\boxed{\Delta y \approx \frac{dy}{dx}\Delta x}$$



This is equivalent to approximating a small segment of a curve with a straight line segment (see figure). This is approximately correct to the degree Δx is small. If Δx is large, error in the estimation will increase.

LINEARIZATION: Approximating a function by it tangent is called linearization. The linearization of $y=f(x)$ at $x=a$ is simply

$$\boxed{y = f(a) + f'(a)(x-a)}$$

This is same as equation of tangent at point $(a, f(a))$.

EXAMPLE: Find linearization of $y = f(x) = \sqrt{1+x}$ at $x=3$.

Solution: $f'(x) = \dfrac{1}{2\sqrt{1+x}} \Rightarrow f'(3) = \dfrac{1}{4}$, $f(3) = 2$ thus

required linearization near $x=3$, $y = f(3) + f'(3)(x-3)$

$\Rightarrow y = 2 + \dfrac{x-3}{4} = \dfrac{x}{4} + \dfrac{5}{4}$, Now this can be used to estimate

$\sqrt{1+x}$ near $x=3$, for instance $\sqrt{4.2}$

$= \sqrt{1+3.2} \approx \frac{3.2}{4} + \frac{5}{4} = 2.050$, while up to four decimals,

it is 2.0494, so our estimate is very close, difference is less than a thousandth.

EXAMPLE: Using Taylor approximation estimate value of $\sin(31^\circ)$

Solution:

$$f(x) \approx f(a) + f'(a)(x-a), \text{ near } 30^\circ,$$

$$y \approx \sin 30^\circ + \cos 30^\circ \times \left(\frac{\pi}{180}\right) \approx 0.5151$$

By Calculator we get $y = 0.5150$ so pretty accurate. Now let us estimate $\sin 36^\circ$,

$$\sin 36^\circ \approx \sin 30^\circ + \cos 30^\circ \times \left(\frac{6\pi}{180}\right) \approx 0.5907$$

By calculator we get $\sin 36^\circ \approx 0.5878$

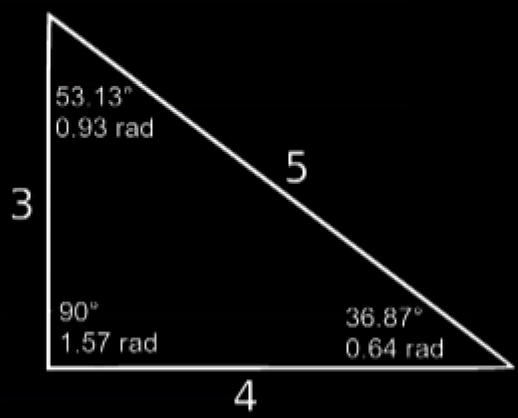
so if we round off to two decimals, our result will be accurate.

→ A special triangle:

$\sin 37^\circ \approx 3/5 \approx \cos 53^\circ$

$\cos 37^\circ \approx 4/5 \approx \sin 53^\circ$

$\tan 37^\circ \approx 3/4 \approx \cot 53^\circ$



EXAMPLE: Assuming $\sin 37^\circ \approx 0.6$ estimate the value of $\sin 40^\circ$.

Solution: $\sin 40^\circ \approx \sin 37^\circ + \cos 37^\circ \times \frac{3\pi}{180} \approx 0.642$

By Calculator, $\sin 40^\circ \approx 0.643$, so again fairly accurate. This technique can be used to estimate a trigonometric ratios of angles near a standard angle.

QUADRATIC APPROXIMATION: Taking up to Δx^2 terms in Taylor expansion gives us Quadratic approximation which is generally more accurate than linear approximation, thus,

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(x-a)^2}{2}$$

EXAMPLE: Find Quadratic approximation of $y = f(x) = \sqrt{1+x}$ near $x = 0$, and hence estimate $\sqrt{1.2}$.

Solution: $f(0) = 1$, $f'(x) = \dfrac{1}{2\sqrt{1+x}} \Rightarrow f'(0) = \dfrac{1}{2}$,

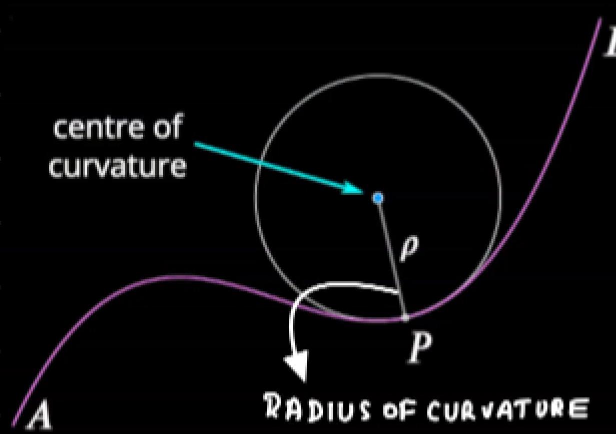
$f''(x) = \dfrac{-1}{4}(x+1)^{-3/2} \Rightarrow f''(0) = \dfrac{-1}{4}$, thus required quadratic approximation:

$$y \approx 1 + \frac{1}{2}x - \frac{1}{8}x^2, \text{ for } \sqrt{1+0.2},\ x = 0.2,$$

$$y \approx 1 + \frac{0.2}{2} - \frac{1}{8} \times 0.04 = 1.095, \text{ By Calculator: } 1.095$$

CENTRE OF CURVATURE AND RADIUS OF CURVATURE:

Any infinitesimal section of a curve can be estimated as arc of a circle (see figure). The radius of curvature of such an arc is called the local radius of curvature of the curve and its centre is called local centre of curvature. The radius of curvature of any curve $y = f(x)$ at a general point is given as:



$$\boxed{R = \frac{(1+y'^2)^{3/2}}{y''}}$$

Thus at a smooth peak or valley, $\boxed{R = 1/y''}$ $(\because y' = 0)$

EXAMPLE: Find radius of curvature of $y = f(x) = x^2$ at $x = 3$ and $x = 0$.

Solution: $y' = 2x$

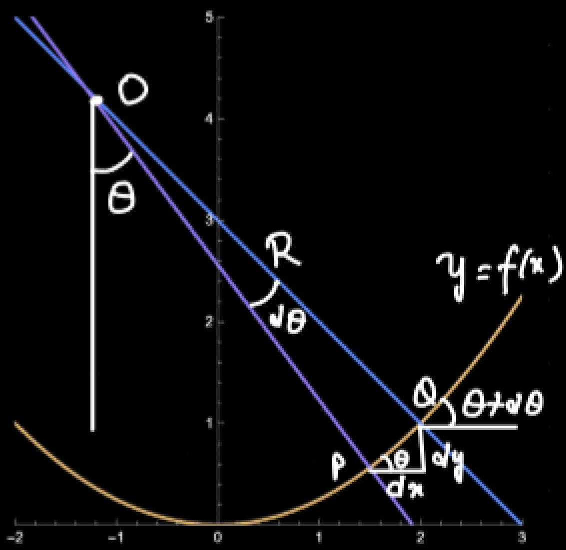
$y'' = 2$

$$\Rightarrow R = \frac{(1+4x^2)^{3/2}}{2} \quad \text{at } x = 3,\ R = \frac{(37)^{3/2}}{2}$$

For $x = 0$, $R = \dfrac{1}{2}$ Ans.

PROOF OF RADIUS OF CURVATURE FORMULA:

Consider the adjacent figure, arc length $\widehat{PQ}$:



$$\widehat{PQ} = R\,d\theta = \sqrt{(dx)^2+(dy)^2}$$

$$\Rightarrow R\,d\theta = dx\sqrt{1+\left(\frac{dy}{dx}\right)^2}$$

$$\Rightarrow R = \frac{dx}{d\theta}\sqrt{1+y'^2} \quad (1)$$

Now $\tan\theta = y'$

$$\Rightarrow \sec^2\theta\,\frac{d\theta}{dx} = y'' \Rightarrow \frac{dx}{d\theta} = \frac{1+\tan^2\theta}{y''}$$

$$\Rightarrow \frac{dx}{d\theta} = \frac{1+y'^2}{y''} \quad (2)$$ Putting (2) in (1):

$$\boxed{R = \frac{(1+y'^2)^{3/2}}{y''}}$$ QED.

EXAMPLE: Find radius of curvature of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at point $(x,y) = (0,-b)$.

Solution: $$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \Rightarrow \frac{2x}{a^2}+\frac{2yy'}{b^2}=0 \quad (1)$$

Now using $x=0$, $y=-b$, $\frac{-2y'}{b}=0 \Rightarrow y'=0$ (2)

Differentiating (1) once more:

$$\frac{2}{a^2}+\frac{2}{b^2}\left[yy''+y'^2\right]=0$$

Now putting $y=-b$, $y'=0$, $\frac{2}{a^2}+\frac{2}{b^2}\left[-by''+0\right]=0$

$\Rightarrow y'' = \frac{b}{a^2}$ (3) Now $R = \frac{(1+y'^2)^{3/2}}{y''} = \frac{1}{b/a^2} = \frac{a^2}{b}$ Ans.